$]-\infty, b[\ (-\infty < x < b)$; $]-\infty, b]\ (-\infty < x \leq b)$

- Let us consider several specific examples of numerical functions. Judging by the appearance of the formulas given below, point out the intervals constituting the domains of the following functions:

$y=\sqrt{1-x^2}$ (1) $y=\sqrt{x-1}$ (2) $y=\sqrt{2-x}$ (3) $y=\frac{1}{\sqrt{x-1}}$ (4) $y=\frac{1}{\sqrt{2-x}}$ (5)

$y=\sqrt{x-1}+\sqrt{2-x}$ (6) $y=\frac{1}{\sqrt{x-1}}+\frac{1}{\sqrt{2-x}}$ (7) $y=\sqrt{2-x}+\frac{1}{\sqrt{x-1}}$ (8)

$y=\sqrt{x-1}+\frac{1}{\sqrt{2-x}}$ (9)

Ans: (1) Interval: $[-1, 1]$ ✓ (2) Interval: $[1, \infty[$ (3) Interval: $]-\infty, 2]$

(4) Interval: $]1, \infty[$ (5) Interval: $]-\infty, 2[$ (6) Interval: $[1, 2]$

(7) Interval: $]1, 2[$ (8) Interval: $]1, 2]$ (9) Interval: $[1, 2[$

Author: Yes, quite right. If a function is a sum (a difference, or a product) of two functions, its domain is represented by the $\boxed{\text{intersection}}$ of the sets which are the domains of the constituent functions.

And what can you say about the domain of the function $\boxed{y=\sqrt{x-2}+\sqrt{1-x}\,?}$

Ans: $\sqrt{x-2}$: $[2, \infty[$ $\sqrt{1-x}$: $]-\infty, 1]$. The intersection is null. The function does not exist.